Bartlett adjustment for random effects meta-analysis of event outcomes with small samples

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Abstract. ... Keywords: Meta-analysis, Bartlett correction, Generalized linear mixed models

1 Introduction

There is currently underway a move toward generalized mixed models in carrying out meta-analyses of event outcomes, particularly in situations where sample sizes are small or events are rare. GLMMs avoid many of the dubious assumptions of the main competing approach, inverse-variance weighting, such as gaussianity of the effects and perfect knowledge of standard deviations. However, inferences under GLMMs are still based on an asymptotic chi-squared approximation. Therefore, as in the case of inverse-variance weighted estimation, the conclusions of smaller meta-analyses are suspect, while smaller meta-analyses are where use of GLMMs is often recommended. A classical remedy is Bartlett correction, which leverages modelling assumptions to improve the efficiency of the testing procedure. When the modelling assumptions are met, the imrpovement is typically a jump in the rate of asymptotic convergence from $1/n$ to $1/n^2$. [[the key modelling assumption in this setting is the gaussianity of the random effect.]]

While Bartlett corrections have long been applied to generalized linear models [[cite]], there do not appear to be extensions to even the simple GLMMs that are used in meta-analysis. The reason for the reluctance appears to be computational difficulty [[cite hanada]]. Unlike in the case of GLMs, closed-form expressions are unavailable for Bartlett corrections for GLMMs. We argue that with careful design these difficulties can be overcome and consumer-grade hardware is sufficient for practical implementation.

In the next section we summarize 4 event data meta-analysis models, following [[Stifjnen's]] exposition. In Section ((ref)) we describe a practical approach to estimation of the Bartlett correction applicfable to these 4 models. In Section ((ref)) we examine the effectiveness and robustness of the corrected test, finding ... . In Section ((ref)) we re-visit the data analyses of ((stijnen)), finding ... .

2 Background

We consider the random effects model for meta-analysis. In this model, a Gaussian random effects models the effect of the individual studies,

$$\theta_i = \mu_0 + \tau_0 Z, \qquad Z \sim N(0,1)$$

The target of inference in the meta-analysis is μ_0, with τ_0^2 a nuisance parameter. The conditional distribution of the data given the random effect depends on the type of analysis being conducted. Following ((stijnen)) we consider 4 types of analyses, determined by whether the studies included are single-arm or two-arm trials, and whether the estimand of the meta-analysis is a proportion or incidence rate.

1. Single arm, proportions. Each study reports the number of events Y_i and total number of trials m_i. The statistical model is binomial

$$f(y_i \mid Z) = \binom{m_i}{y_i} p^{y_i}(1-p)^{m_i - y_i}$$
$$p = \text{expit}(\mu_0 + \tau_0 Z_i)$$

   The target of inference μ_0 represents the log-odds of an event. The number of trials m_i is treated as a parameter of the conditional PMF $f_i = f_{m_i}$, which therefore depends on the study in the meta-analysis.[[replace f with fi in displays, update next 3 similarly]]

2. Two arms, proportions. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the total number of trials $m_i^{(T)}$ and $m_i^{(C)}$. The statistical model conditions on the total number of events $S_i = Y_i^{(T)} + Y_i^{(C)}$, under which $Y_i^{(T)}$ follows Fisher's non-central hypergeometric distribution

$$f(y_i^{(T)} \mid Z_i) = \frac{\binom{m_i^{(T)}}{y_i^{(T)}}\binom{m_i^{(C)}}{S_i - y_i^{(T)}}\omega^{y_i^{(T)}}}{\sum_{\max(0, S_i - m_i^{(C)})}^{\min(S_i, m_i^{(T)})} \binom{m_i^{(T)}}{y}\binom{m_i^{(C)}}{S_i - y}\omega^{y}}$$
$$\omega = \exp(\mu_0 + \tau_0 Z_i)$$

   The target of inference μ_0 represents the ratio of the treatment to control log-odds. The parameters of the conditional PMF include $S, m^{(T)}, m^{(C)}$.[[ref to bivriate model which models baseline risk, mentions assumptions in this approach]]

3. Single arm, incidence rate. Each study reports the number of events Y_i and the duration of observation T_i. The model is Poisson,

$$f(y_i \mid Z_i) = \exp(-\lambda)\frac{\lambda^{y_i}}{y_i!}$$
$$\lambda = T_i \exp(\mu_0 + \tau_0 Z_i)$$

   The target of inference represents the log of the incidence rate. The follow-up time T_i is treated as non-random.

4. Two arms, incidence rates. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the duration of observation $T_i^{(T)}$ and $T_i^{(C)}$. Conditionally on $S_i = Y_i^{(T)} + Y_i^{(C)}$, $Y_i^{(T)}$ is binomial

$$f(y_i^{(T)} \mid Z_i) = \binom{S_i}{y_i^{(T)}} p^{y_i^{(T)}} (1-p)^{S_i - y_i^{(T)}}$$
$$p = \text{expit}(\log(T_i^{(T)}/T_i^{(C)}) + \mu_0 + \tau_0 Z_i)$$

The target of inference is the logarithm of the ratio of the incidence rates between the two arms. As before the periods $T_i^{(T)}$ and $T_i^{(C)}$ are teated as parameters of the conditional PMF.

In all cases the likelihood for study i has the form:

$$L_i(\mu, \tau^2 | Y_i) = \int_{-\infty}^{\infty} f_i(Y_i|z)\phi(z)dz$$

where $f_i(y|z)$ is study i's PMF conditional on a Gaussian random effect, as given above.

The likelihood ratio test of the null $\mu = \mu_0$ is based on the test statistic

$$W = 2\sum_{i=1}^{n} (\log L_i(\hat{\mu}, \hat{\tau}^2|Y_i) - \log L_i(\mu_0, \hat{\tau}_0^2|Y_i))$$

[[subscript on i to transmit dependence of pmf f on i]] where $(\hat{\mu}, \hat{\tau}^2)$ maximizes $(\mu, \tau^2) \mapsto \sum_{i=1}^n \log L_i(\mu, \tau^2|Y_i)$ and $\hat{\tau}_0^2 = \hat{\tau}_0^2(\mu)$ maximizes $\tau^2 \mapsto \sum_{i=1}^n \log L_i(\mu_0, \tau^2|Y_i)$. In routine applications W converges to a chi squared distribution under the null hypothesis, by Wilks' Theorem. Relying on this asymptotic approximation a confidence interval for μ_0 can be formed by inverting the test statistic.

Under a range of conditions [[citations]] the mean of a correctly specified LR statistic ((ref)) admits an expansion in integral powers of $1/n$

$$\mathbb{E}(W) = 1 + b/n + O(1/n^2).$$

Therefore, the Bartlett-corrected statistic

$$W_B = \frac{W}{1 + b/n}$$

is de-biased to order $O(1/n^2)$. The work of ((lawley and bartlett)) shows further that, under appropriate conditions, not only does the mean of ((ref above)) match the mean of a chi squared to order $O(1/n^2)$, but also all other moments, and so the entire distribution. Then the resulting asymptotic rate of convergence to a chi squared random variable is $O(1/n^2)$. As with ((ref lr statistic)), the adjusted statistic may be inverted to obtain CIs and p-values. [[mention C depends on parameters, or maybe make explicit in the notation]]

Among the assumptions underlying the above development we focus on two prominent ones. The first is correct specification of the likelihood, in particular, gaussianity of the unobserved random effect. [[we refer to sitjnen to other modeling assumptions implicit in

((ref 1–4 above)), such as ... .]] While the LR test itself depends on gaussianity for valid inference ((citation showing effects of bad estimatio of tau2)), the Bartlett correction doubles down on the assumption. Second, the arguments that the Bartlett correction to deliver $1/n^2$ rate are usually justified by Cramér's condition for the underlying random variables, which excludes lattice variables such as are considered here. while bartlett correction has been applied to lattice random variables since at least the 80s, their effectiveness does not appear to be understood in much generality ((cite frydenberg re multinomial ngative result and gotze, and janson re positive result)). [[of bartlett correction does not open question since the 80s, when lattice RVs can be bartlett codnitions. cramers condition has been relaxed in recent work ((cite)) but these conditions are difficult to evaluate in specific cases. they do offer promise that the entireprise of bartlett correcting count data, which has been done since 80s ((cite cordeiro)), is not pointless.]] we have no theoretical contibutions to make here and will examine both model robustness and discreteness by simulation.

3 Method

lawley ((cite)) provides expressions for the correction ((ref)) as sums and products of the components of the inverse Fisher information and certain tensors [[k terms]]. the specific formulas arent of interest here; being a small set of sums and products they don't introduce numerical or computational difficulties that aren't already to be found in the basic k terms. Let D_r denote partial differentiation with respect to a sequence of the parameters μ and τ^2 given by r. For example, ... denotes the off-diagonal term of the obseerved information matrix for ((ref total likelihood)) Define $k_r = \mathbb{E} D_r l(\mu, \tau^2)$ and $k_r^{(s)} = D_s k_r$. In this notation the Fisher information matrix has components k_{11}, k_{12}, k_{22}.[[intro component notation or use mu,tau2 here?]]. Lawley's formulas for computation of the correction require all terms k_r where r is of length 2, 3, or 4, and all terms $k_r^{(s)}$ where r is of length 3 and s of length 1 or r is of length 2 and s of length 1 or 2. Taking into account permutation symmetry of the order of differentiation, there are 35 such terms.

To compute the tensor components ((ref)) for the likelihood ((ref)), we approximate the integral by a quadrature rule. Since all the likelihoods ((ref)) involve integrating against a Gaussian, Gauss-Hermite quadrature is a natural choice. This and similar methods consist in a weights and nodes in the domain of integration, the size of which controls the quality of approximation, allowing the integral ((ref)) to be approximated by a weighted sum on, say ((how much detail to give here,[[set notation nGH and w. this tuning parameter will be needed later. say we are using fixed rule not adatpve]]. maybe just leave it as a sum approximation.))

$$L(\theta, \tau^2|Y) = \int_{-\infty}^{\infty} f(Y|z)\phi(z)dz \approx \sum_j w_j f(Y|z_j)$$

A higher-order partial derivative of the log-likelihood may be approximated by $D_r \log L(\mu, \tau^2|Y) \approx D_r \log \sum_j w_j f(Y|z_j)$. The expectation of the latter may be approximated by another quadra-

ture

$$
\begin{aligned}
k_r &= \sum_y \left(D_r \log L(\mu, \tau^2|y)\right) \int f(y|z)\phi(z)dz \\
&\approx \sum_y \left(D_r \log L(\mu, \tau^2|y)\right) \sum_k w_k f(y|z_k) \\
&\approx \sum_y \left(D_r \log \sum_j w_j f(y|z_j)\right) \sum_k w_k f(y|z_k)
\end{aligned}
$$

The outer sum is taken over the support of Y, which is a finite set in all the models considered here other than the Poisson model. The Poisson model requires a truncation rule, discussed further below. The general tensor component needed for the computation of the Bartlett correction factor is then

$$
k_r^{(s)} \approx \sum_y D_s \left(\left(D_r \log \sum_j w_j f(y|z_j)\right) \sum_k w_k f(y|z_k) \right)
$$

Carrying these differentiations ((ref above display)) through yields rational expressions in derivatives of the weighted sums $\sum_j w_j f(y|z_j)$. The form of these 35 expressions, for the 35 k terms, are the same for any model based on a likelihood of the form ((ref)), and may be pre-computed. For example $k_{\mu,\mu} = F_{20} - (F_{10}^2/F_{00})$, and $k_{\mu,\tau^2}^{(\tau^2)} = (F_{01}^2 F_{10})/F_{00}^2 - (F_{02}F_{10} + F_{01}F_{11})/F_{00} + F_{12}$, with the notation $F_{jk} = \sum_j w_j \frac{d^j}{d\mu^j} \frac{d^k}{d(\tau^2)^k} f(y|z_j)$.[[notatoin clash – j]]

The derivatives F_{jk} of the model-specific conditional PMFs that appear in the 35 tensor components number 15: F_{jk} for $0 \leq j + k \leq 4$. The form of these derivatives may be derived independently of data but their specific values in general depend on parameters of the conditional PMF ((see above)), and must typically be computed for each study of a meta-analysis. The latter computations may be carried out in parallel when the meta-analysis consists of stochatically independent studies, such as we consider here. Using well-known recurrences ((cite johnson and kotz univariate discrete distributions)) the derivatives $F_{jk}, 0 \leq j + k \leq 4$, may be re-written as polynomial expressions of $F_{00} = \sum_j w_j f(y|\theta_j)$,[[add i subscript to f's? also in derivation above??]] for which optimized and numerically stable routines are available. For example, for the model ((ref poisson/incidence rate model)),

$$
\frac{d}{d\lambda_i} f_{\lambda_i}(y_i \mid Z_i) = f_{\lambda_i}(y_i - 1 \mid Z_i) - f_{\lambda_i}(y_i \mid Z_i)
$$

[[make Fjk functions 0f y, so this recurrence can be written in terms o fthem]]

mention CAS, andrews and stafford, old lawley program.

further implementation details. support for poisson. speed issues with hyperg.

4 Simulation

we examine the rate of convergence ot a chi squared of the LR and Bartlett-corrected LR. rate approximated using log-log. data parameters are sampled from stijnen tables ((data analysis)) so as to resemble real data. we also examine robustness of the gaussianity assumption.

since the leading term in CLT-type approximations is based on the skew, we examine how the rate varies as the skew of the underlying RV is varied. we use the sinh-asinch model ... to obtain a family of RVs based on the Gaussian with the skew perturbed.

5 Data analysis

We revisit the data analysis of ((stijnen2010)).

4a. proportions

4b. incidence